

2(a)	<i>any two from:</i> ('kinks') prevents close packing of, phospholipids / membrane components, (at low temperature) ; keeps / maintains, fluidity ; A increases A ora - prevents becoming too rigid <i>idea of</i> preventing damage to membranes by preventing freezing ; maintains movement of (named) substances across membranes ; A any named example of movement across membrane R increases AVP ; e.g. maintains movement of proteins within membrane	2
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Question	Answer	Marks
2(b)(i)	<i>idea that triglycerides and phospholipids</i> are not composed of, monomers / repeating (sub-)units ;	1
2(b)(ii)	<i>any two from:</i> <i>phospholipids have</i> two fatty acids (residues / tails) not three ; A hydrocarbon chains / aliphatic chains A one less fatty acid (residue / tail) A ora two ester bonds rather than three ; a phosphate (group / head) ; A ora R 'phosphate not glycerol' AVP ; e.g. may have a (named) additional group, such as choline / AW e.g. triglycerides do not have nitrogen / phospholipids may have nitrogen	2
2(c)	<i>any four from:</i> 1 thromboxane is a (cell) signalling molecule ; A 'thromboxane acts as a signal' I 'messenger' / hormone 2 released into / circulates in / AW, blood / plasma ; 3 (smooth) muscle, cell / tissue, is <u>target</u> ; 4 thromboxane binds to <u>receptors</u> ; 5 <i>ref. to</i> thromboxane is complementary to receptor ; 6 (specific) response is smooth muscle (cell) contraction ; R smooth muscle constricts 7 AVP ; e.g. detail of change, such as activating G proteins / secondary messenger / enzyme cascade / chain of reactions / AW	4